

Digital Logic

Day 7

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Delay

We need more than just logic in order to make a computer.

The Delay component outputs whatever was its input on the previous simulation step.

We now have enough to make a computer!



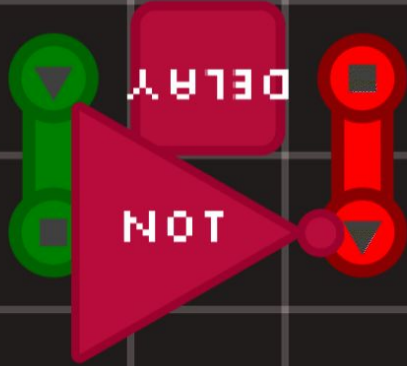
The delay component outputs **OFF** on its first tick



You can slow down a signal by chaining delay components.

Your turn - make a wire blink.

It should turn on and off every other tick. Use one 1 delay component.



With delay components we can store a wire's power for a single tick. We will use this to create computer memory.

What operations would memory need to do?

Reading

Writing

Memory Cell

Create a new component.

This will be a 1 bit memory cell.

IN0 is *whether or not* to write IN1 into the memory cell.

IN1 is the value to write into the memory cell.

MEM1

16

16

IN 0



IN 1



MEM1

OUT 0



The output is the value currently stored in the memory cell.

Memory Cell

Tick	IN1 (Written value)	IN2 (Enable Write)	Description	OUT1
1	1	1	Write 1 into memory cell.	0
2	0	1	Keep stored value.	1
3	0	0	Keep stored value.	1
4	1	0	Write 0 into memory cell.	1
5	0	1	Keep stored value.	0